



Research article

Temporal molecular remodeling of T cells informs their possible adaptation in 4T1 tumors

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ABSTRACT

Background: The triple-negative breast cancer (TNBC) microenvironment undergoes progressive reprogramming, transitioning from an early immune-active state to a late immune-suppressed state. While tumor cell plasticity has been extensively studied, the temporal molecular remodeling of T cells *in vivo* remains poorly defined.

Results: Transcriptional analysis of T cells within 4T1 TNBC tumors, harvested at one-, three-, and six-weeks post-tumor implantation in the mammary fat pads of BALB/c mice, revealed a decline in transcriptomic signatures associated with T cells from 194 at one week to 156 at six weeks, with a significant late-stage loss or reduction of transcripts related to T cell receptors (TCR), natural killer T, and gamma delta T cells. Furthermore, changes in various temporal signature genes specific to T cell cytokines and transcription factors reflected temporal T cell polarization to CD4⁺ type 1 T helper and type 1 CD8⁺ cytotoxic T cell responses at one week, CD8⁺ cytotoxic follicular T cell skewing at three weeks, and interleukin 17/22 producing CD8⁺ T cell transition at six weeks. The antigen-presenting cell (APC) transcripts deteriorated at six weeks, characterized by reduced expression of co-stimulatory and APC genes. Despite an early dominance of M1-like macrophage genes (e.g., *IL-12α/β*), persistent expression of arginase 1 (*ARG1*) and other M2-associated genes indicated a stable tolerogenic niche.

Conclusions: The temporally coordinated immune shifts, such as progressive decline in transcripts associated with T cell functions, TCRs, APCs, and sustained macrophage-driven immunosuppression, suggest tumor-driven adaptation toward immune evasion and identify potential windows for stage-specific immunotherapeutic intervention.

1. Introduction

Triple-negative breast cancer (TNBC), defined by the absence of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2), represents one of the most clinically challenging subtypes of breast cancer [1]. This receptor-deficient profile renders TNBC unresponsive to endocrine or HER2-directed therapies. TNBC is particularly aggressive, with high recurrence rates and poor prognosis, especially in the metastatic setting [2]. Immunotherapies,

such as immune checkpoint blockade, antibody-drug conjugates, oncolytic viruses, and cancer vaccines, are emerging treatment options for TNBC. However, their efficacy relies strongly on the functional state of the host immune system, particularly tumor-infiltrating T lymphocytes (TILs) [3–5]. Within the tumor microenvironment (TME), TILs encounter strong immunosuppressive cues that limit their ability to mount effective antitumor responses [6].

T cells show a remarkable degree of molecular plasticity, undergoing dynamic and reversible changes in phenotype, function, gene

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expression, and metabolism in response to extrinsic stimuli [7, 8]. This adaptability is fundamental for immune surveillance, enabling T cells to recognize diverse pathogens and function across tissue contexts. In cancer, however, the TME can hijack this T cell plasticity to promote immune evasion. For instance, cytotoxic CD8⁺ T cells often become dysfunctional or exhausted due to chronic antigen exposure, signaling related to inhibitory ligands such as programmed death-ligand 1 (PD-L1), and nutrient depletion [9, 10]. Likewise, CD4⁺ helper T cells may be skewed toward regulatory T cells (Tregs) under the influence of transforming growth factor beta (TGF- β) and interleukin 10 (IL-10), thereby dampening effective immune surveillance and antitumor immunity [11, 12].

T cells also shape tumor evolution [13]. For instance, interferon-gamma (IFN- γ), secreted by activated T cells (and other immune cells), is traditionally viewed as an antitumor cytokine [14]; however, sustained IFN- γ signaling can paradoxically promote tumor cell survival, epithelial-to-mesenchymal transition (EMT), and metastasis [15–17]. This bidirectional relationship highlights that T cell plasticity not only reflects immune dysfunction but also drives tumor adaptation. Therapies that limit harmful plasticity (e.g., preventing T cell exhaustion or Treg conversion) represent promising avenues to improve immunotherapy outcomes [18]. Despite increasing recognition of T cell plasticity, its temporal evolution within the TME remains poorly understood. A better understanding of the temporal remodeling of T cells in tumors could shed light on how tumors evade immune recognition and how immune cells might be reprogrammed to overcome these barriers. Previously, we investigated the molecular plasticity of tumor cells in the 4T1 TNBC model system [19], a well-characterized murine TNBC model that recapitulates key histopathological and immunological features of human TNBC. Building on that work, the present study investigates the molecular remodeling of T cells during 4T1 tumor progression *in vivo*. Using longitudinal transcriptomic analyses of 4T1 tumors collected at early, intermediate, and late stages of growth, we examined how evolving T cell states reflect and influence tumor adaptation, progression, and immune evasion. Our findings provide a dynamic framework for understanding temporal T cell remodeling in TNBC, highlighting potential windows for immunotherapeutic intervention.

2. Methods

To assess temporal remodeling of T cells in 4T1 bulk tumors over time *in vivo*, we retrieved paired-end RNA-seq sequencing data from nine samples belonging to the NCBI BioProject PRJNA588756, which included bulk tumor samples derived from the same batch of 4T1 cells implanted orthotopically into the mammary ducts of BALB/c mice, collected at one week (early), three weeks (intermediate), and six weeks (late) post-tumor implantation [20]. We additionally used 4T1 cells, which are syngeneic to BALB/c mice, as a baseline in this study and obtained them from NCBI BioProjects PRJNA671832, PRJNA873199, and PRJEB36287 (Table S1).

Transcriptomic sequence reads were quality-checked with FastQC v.0.12.1 [21], subsequently trimmed using Fastp v.0.24.0 [22], and aligned to the GRCh39 mouse reference genome (Ensembl release 115) using HISAT2 [23]. Sequencing depth was between 7.1 and 10.5 million paired-end reads per sample, with high mapping rates to the reference genome (90.4–94.6 %) (Table S2). Gene-level read counts were quantified with FeatureCounts [24]. To test for differences in expression between baseline and each of the three time points, differential expression analysis was conducted in the R statistical program (v. 4.5.1) using the R package DESeq2 [25]. After sequence count normalization, differentially expressed genes (DEGs) were identified based on an adjusted *p*-value < 0.05 and an absolute log₂-fold change (log₂FC) $\geq$ 1. Upregulated and downregulated genes were classified using log₂FC $\geq$ 1 and log₂FC $\leq$ -1, respectively. Default parameters were used for all aforementioned software. The log₂FC values, unless otherwise specified,

were indicated in parentheses in the Results and Discussion sections.

DEGs were filtered to include only T cell-related genes, identified using the nCounter Mouse PanCancer Immune Profiling Panel [26] and cross-referenced with published literature to ensure biological relevance. The resulting DEGs were visualized in R using volcano plots and heatmaps [27]. We applied the BASE algorithm as a deconvolution method to estimate immune cell composition, as presented in Table S3, using the immunedeconv R package [28]. Because this study uses bulk RNA-seq, the results reflect population-level transcriptional correlations rather than direct functional interactions between specific cell types. For statistical comparisons, one-way analysis of variance (ANOVA) followed by the Tukey test was performed using GraphPad Prism v10.5.

3. Results

3.1. Dynamic molecular T cell profiles across stages of 4T1 tumor growth

Tumor plasticity not only reshapes cancer cells but also drives dynamic remodeling of the tumor immune microenvironment, including T cell compartments [13]. To elucidate the temporal remodeling of T cell compartments, we profiled key T cell-associated genes during 4T1 tumor progression from one to six weeks. Out of 200 T cell-specific genes outlined in the Mouse PanCancer Immune Profiling Panel [26], 194 genes were expressed in one-week tumors, decreasing to 191 genes at three weeks and further to 156 genes at six weeks (Fig. 1). This gradual reduction suggests that T cell functionality and activity undergo dynamic modulation as tumors grow, reflecting a complex interplay between tumor growth and immune regulation.

To characterize the molecular transcriptomic signatures associated with NKT cell function, we profiled 33 key NKT-associated genes [26] at one-, three-, and six-weeks post-tumor implantation. Between one and three weeks, high levels of expression for NKT-associated genes, such as *perforin 1* (*PRF1*; 12.44), *granzyme B* (*GZMB*; 13.06), *interferon-gamma* (IFN- γ ; 6.89), *interleukin-13* (*IL-13*; 9.76), and *interleukin-21 receptor* (*IL-21R*; 11.90), indicate early emergence of regulatory and innate-like profiles as well as a robust NKT effector phenotype. The high log₂FC values (>10) of NKT-associated genes are likely a consequence of using 4T1 tumor cells as the baseline, which could lead to fold-change inflation. By six weeks, cytokine genes such as *IL-13* (5.67) were sharply downregulated, while *IL-21* was lost, and transcriptomic expression of cytotoxic markers *PRF1* and *GZMB* persisted (Tables 1 and 2, and S4). These results may reflect changes in NKT cell infiltration at different stages of tumor development; however, statistical analysis involving NKT cell-related genes revealed no significant differences at six weeks compared to one and three weeks (Fig. 2A).

TCR-related variable (V) and joining (J) segment genes, including TCR beta variable 2 (*TRBV2*), 16 (*TRBV16*) and 17 (*TRBV17*), as well as TCR beta joining 1–1 to 1–5 (*TRBJ1–1* to *TRBJ1–5*) and 2–1 (*TRBJ2–1*), became undetectable by six weeks (Table S5). No significant difference in TCR-related gene expression was observed between one and three weeks (*p* = 0.3159), suggesting stable TCR gene expression during the early to intermediate phases of tumor growth. Interestingly, at six weeks (*p* < 0.01), these genes were significantly decreased, suggesting a substantial reorganization of the TCR transcriptome by six weeks post-implantation. Conversely, TCR alpha variable 1 (*TRAV1*) was absent in the early and intermediate phases of tumor growth, then emerged at the late stage. Likewise, TCR alpha variable 3–3 (*TRAV3–3*) was not detected in early stage but emerged at intermediate stage and persisted through the late stage of tumor growth (Fig. 2B). Overall, loss of TCR-related variable (V) and joining (J) segment genes at six weeks and the emergence of TCR alpha variable genes at six weeks suggests a compensatory mechanism to sustain immune activity within a clonally restricted population.

The temporal plasticity of gamma delta ($\gamma\delta$) T cells in the TME offers important insights for cancer immunotherapy [29]. In our study, $\gamma\delta$ T cell-associated genes displayed no significant transcriptional shifts at six

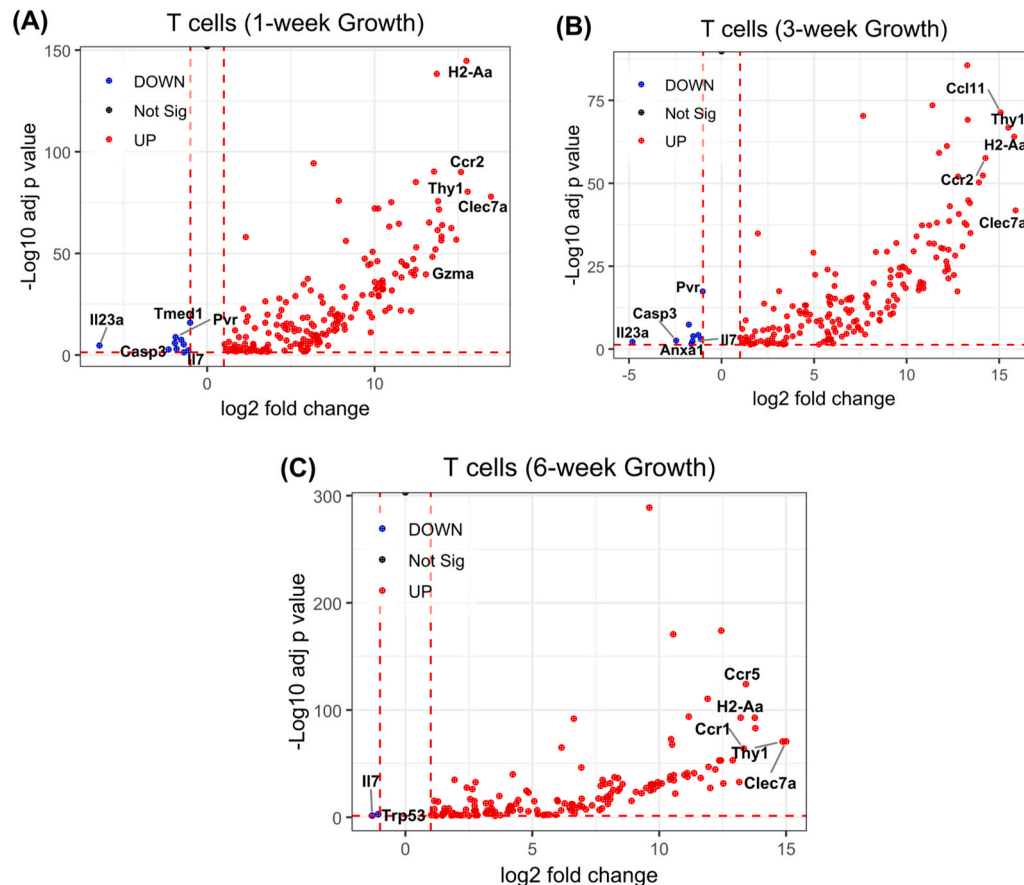


Fig. 1. Volcano plots for DEGs that are T cell-associated genes, as determined by the Mouse PanCancer Immune Profiling Panel [26], at different stages of 4T1 tumor growth. Compared to the 4T1 cell RNA-seq dataset as a baseline, these plots show gene expression changes at (A) 1 week (194 genes), (B) 3 weeks (191 genes), and (C) 6 weeks (156 genes) post-tumor implantation. The X-axis (log2FC) indicates the magnitude and direction of expression changes, with positive values denoting higher expression and negative values denoting lower expression, while the Y-axis (-Log10 adj p value) indicates the strength of the statistical significance of these expression changes.

The top 5 upregulated and the top 5 downregulated genes are labeled in each plot.

weeks compared to earlier time points (Fig. 2C). Several $\gamma\delta$ TCR genes, such as TCR gamma variable 1 (*TRGV1*) and 6 (*TRGV6*), TCR delta variable 4 (*TRDV4*) and 5 (*TRDV5*), and SRY-box transcription factor 13 (*SOX13*) were lost (Fig. 2C, Table S6). However, cytotoxic mediator genes *PRF1* and *GZMB* remained relatively stable or slightly elevated, while *IFN- γ* peaked transiently at three weeks (8.53) before declining at six weeks (6.58) (Fig. 2D, Table 1 and S6). T cell activation genes, such as cluster of differentiation 69 (*CD69*), partially declined from 10.08 (earlier) to 9.49 (late stage) (Table 1). The cluster of differentiation 3 isoforms (*CD3E*, *CD3 δ* , and *CD3 γ*) also declined by late stage, indicating weakened TCR signaling (Fig. 2E, Table 1). *CD4* expression declined sharply at six weeks (10.99–7.69), and *CD8 α/β 1* peaked at one to three weeks before decreasing at six weeks (Fig. 2F–G, Table 1), reflecting early activation followed by late-stage dysfunction.

3.2. Longitudinal shifts in T cell subsets during tumor progression

Cytokines and transcription factors orchestrate T cell polarization over time [30]. For example, at one week, downregulation of *IL-12 α* , *IL-12 β* ($p < 0.01$), and signal transducer and activator of transcription 4 (*STAT4*; $p < 0.01$) (Table S7) indicates polarization of T cells to type 1 $CD4^+$ T helper and type 1 $CD8^+$ cytotoxic T cells (Th1/Tc1) [31]. At three weeks, transient but significant ($p < 0.05$ vs. one week) increases in *IL-21* suggest $CD8^+$ follicular cytotoxic T cell (Tfc)-like response [32], while modest yet significant ($p < 0.05$) increases in *IL-1 β* indicate emergence of *IL-17/IL-22*-producing $CD8^+$ Tc17/Tc22-like cells [33–35]

(Fig. 3A, Table S7). By six weeks, significant ($p < 0.01$) induction of aryl hydrocarbon receptor (*AhR*) ($p < 0.05$) and increased *IL-1 β* expression is an indication of Tc17/Tc22 polarization [36] (Fig. 3A–B, Table S7). In contrast, a significant ($p < 0.05$) decline of forkhead box P3 (*FOXP3*) and *STAT5 α* at six weeks indicates functional loss of $FOXP3^+CD8^+$ Tregs [37] (Fig. 3B, Table S7).

3.3. T cell-related cytokine and chemokine remodeling during tumor growth

Pro-inflammatory cytokines were dynamically regulated (Fig. 4A, Table 2). For instance, *IFN- γ* , *GZMB*, and *PRF1* expression remained elevated across all time points, suggesting Th1/Tc1 activity [38, 39]. Th2 cytokines *IL-5* and *IL-13* were downregulated at six weeks (*IL-5*: 5.88–0; *IL-13*: 10.69–5.67; $p < 0.01$ vs. three weeks) (Table 2). The expression of *IL-21*, a cytokine gene implicated in $CD8^+$ Tfc-like responses, peaked at three weeks but was lost by six weeks (Table 2), suggesting transient $CD8^+$ Tfc activation that is often associated with B cell help and early tumor immune surveillance [40].

Parallel to pro-inflammatory cytokines, the immunosuppressive cytokine-related genes appeared to undergo substantial remodeling. For example, *IL-10* (5.88 at three weeks to 3.53 at six weeks) and indoleamine 2,3-dioxygenase 1 (*IDO1*; 6.78–4.53) decreased over time, whereas arginase-1 (*ARG1*) remained high (9.4 at one week to 9.14 at three weeks to 9.61 at six weeks) (Fig. 4B, Table 2). Chemokine profiling offers additional insight into the temporal dynamics of immune cell

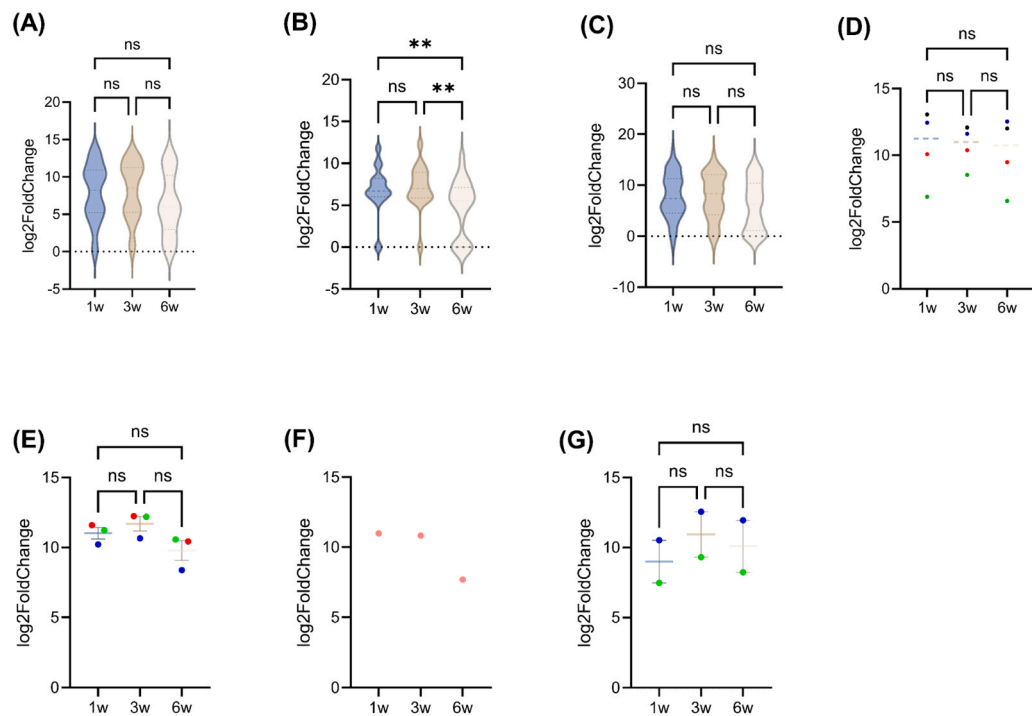


Fig. 2. T cell expression profiles across different stages of tumor growth. Log2FC of various genes related to (A) natural killer T (NKT) cells, (B) T cell receptor (TCR), (C) gamma delta ($\gamma\delta$) T cells, (D) T cell activation markers (CD69 in red, IFN γ in green, PRF1 in blue, and GZMB in black), (E) CD3 isoforms (CD3 ϵ in red, CD3 γ in green, and CD3 δ in blue), (F) CD4, and (G) CD8 isoforms (CD8 α in blue and CD8 β 1 in green). ns, not significant; w, weeks; ** $p \leq 0.01$. Figure A-C Genes presented in these figures are exactly reflected in [Tables S4-S6](#).

Table 1
Log2FC of T cell lineage and their activation markers across different stages of tumor growth.

Gene	1 week	3 weeks	6 weeks
<i>Activation markers</i>			
CD69	10.08	10.38	9.49
PRF1	12.44	11.60	12.53
IFN- γ	6.89	8.53	6.58
GZMB	13.06	12.08	12.01
<i>CD3 Isoforms</i>			
CD3 ϵ	11.60	12.24	10.43
CD3 δ	10.22	10.65	8.38
CD3 γ	11.23	12.20	10.57
<i>T cell subsets</i>			
CD4	10.99	10.83	7.69
CD8 α	10.52	12.55	11.94
CD8 β 1	7.47	9.31	8.23

trafficking. For instance, C-X-C motif chemokine ligand 9 (CXCL9), a potent chemoattractant for CXCR3⁺CD8⁺ T cells and CD4⁺ Th1 cells [41], remained consistently elevated over time (10.22–11.22; $p < 0.05$), suggesting sustained recruitment of cytotoxic effectors to the TME. CXCL10 levels increased linearly over time ($p < 0.05$), peaking at six weeks (3.17), which reinforces the notion of increasing effector T cell infiltration during tumor progression (Table 2). Conversely, Treg-attracting regulatory chemokines, such as CCL17 and CCL22, were significantly downregulated over time (CCL17: 7.74–3.40; CCL22: 13.45–9.69; $p < 0.01$). In contrast, changes in transcriptomic signatures associated with immunosuppressive cytokines and chemoattractants were not significant (Fig. 4A–C, Table 2).

3.4. Stable inhibitory checkpoint expression with progressive loss of co-stimulation

Recruited T cells within the TME often exhibit features of functional

exhaustion due to chronic antigen stimulation and persistent exposure to immunosuppressive cues [42, 43]. This exhausted phenotype is typically marked by sustained expression of multiple inhibitory immune checkpoints, such as programmed cell death 1 (PD-1), T-cell immunoglobulin and mucin domain-containing molecule 3 (TIM-3), lymphocyte activation 3 (LAG-3), and cytotoxic T lymphocyte antigen 4 (CTLA-4), that collectively dampen T cell effector functions and promote immune tolerance [44]. Recent evidence, however, highlights the phenotypic plasticity of these exhausted T cells, indicating that exhaustion exists on a continuum rather than as a terminal state [45].

In this study, transcriptomic signatures associated with exhaustion marker genes PD-1, CTLA-4, LAG3, T-cell immunoreceptor with immunoglobulin and ITIM domains (TIGIT), and B and T lymphocyte attenuator (BTLA) remained stable over six weeks ($p = 0.9$ vs. three weeks) (Fig. 5A, Table 3). In contrast, transcriptomic signatures associated with T cell co-stimulatory molecules CD28 and ICOS were significantly downregulated at six weeks compared to the one- and three-week time points ($p < 0.01$ and $p < 0.001$, respectively) (Fig. 5B, Table 3), indicating progressive functional compromise of TILs. These findings support a model in which TILs become increasingly entrenched in a functionally compromised state, marked by stable expression of inhibitory receptors and diminished co-stimulatory signaling, which are hallmarks of terminal exhaustion [46, 47].

3.5. Changes in transcriptional dynamics of APCs over time

Antigen-presenting cells (APCs), such as dendritic cells (DCs), serve as the primary sentinels of the immune system by capturing debris from dying tumor cells, processing these antigens, and presenting them on MHC class I or II molecules to activate CD8⁺ cytotoxic or CD4⁺ helper T cells, respectively [48]. The functionality of early-stage APCs was evident from high expression of APC-associated transcriptomic signatures, C-C chemokine receptor type 7 (CCR7; 7.347), which is essential for APC migration to draining lymph nodes [49], and MHC Class II genes H2-Aa (15.485) and H2-Ab1 (10.636) at one week (Table S8). By six

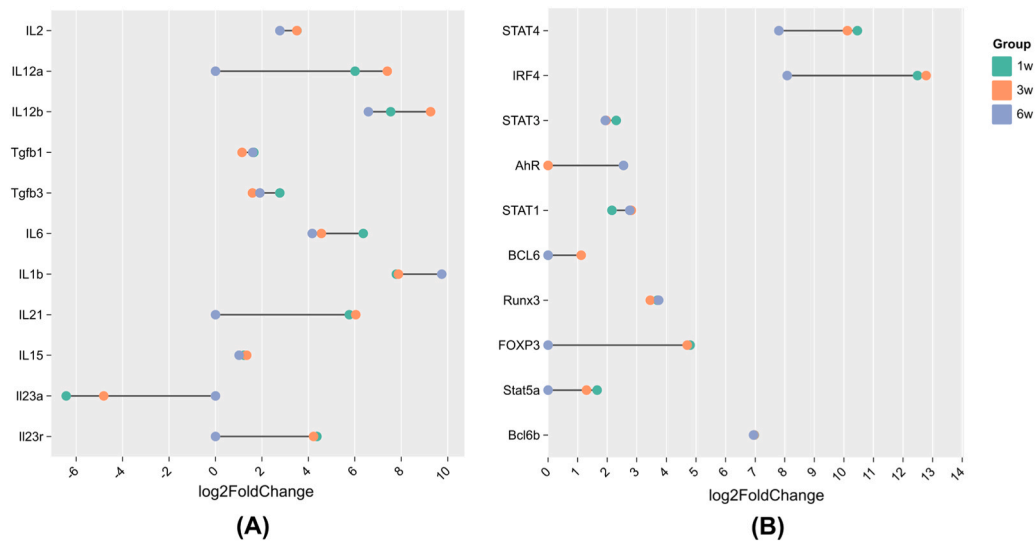


Fig. 3. Expression patterns of T cell-related genes associated with (A) polarization and (B) transcription. A list of genes related to T cell polarization and transcription can be found in Koh et al. 2023 [30].

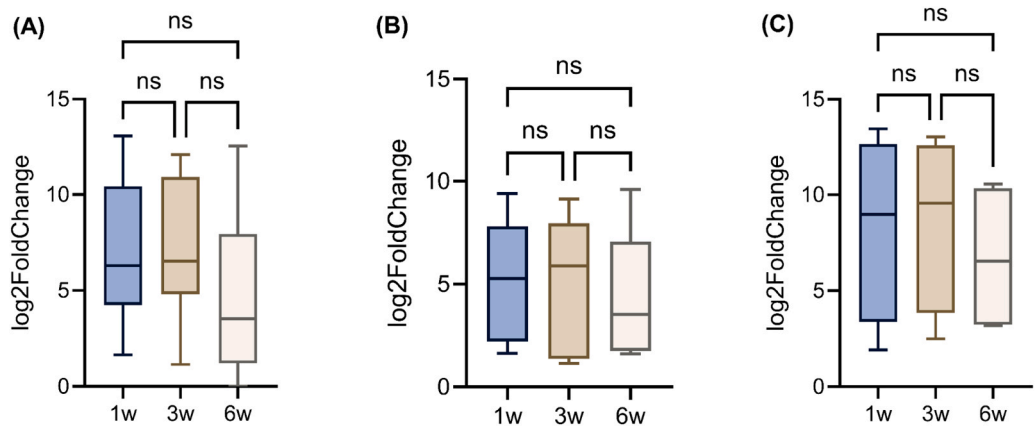


Fig. 4. Log2FC of genes related to (A) pro-inflammatory cytokines, (B) immunosuppressive cytokines, and (C) chemoattractants. ns, not significant; w, weeks. Genes presented in these figures are reflected in Table 2.

Table 2
Log2FC of T cell-associated proinflammatory and immunosuppressive cytokines and chemoattractants.

Gene	1 week	3 weeks	6 weeks
<i>Proinflammatory cytokines</i>			
IL5	4.741	5.882	0
IL13	9.767	10.685	5.674
IL9R	6.35	7.02	3.538
IL21	5.76	6.04	0
<i>Immunosuppressive cytokines</i>			
IL10	6.243	5.879	3.529
TGFβ1	1.65	1.143	1.6038
TGFβ3	2.772	1.599	1.91
IDO1	5.273	6.78	4.527
ARG1	9.406	9.14	9.611
<i>Chemoattractants</i>			
CXCL9	10.221	11.218	10.551
CXCL10	1.9208	2.4938	3.171
CCL17	7.738	7.904	3.395
CCL22	13.453	13.02	9.693

weeks, the expression of other APC-related genes, including *CD1d1*, *H2T23*, and *ICAM1*, had declined or was lost, indicating a compromise of classical antigen-presenting function (Fig. 6A, Table S8).

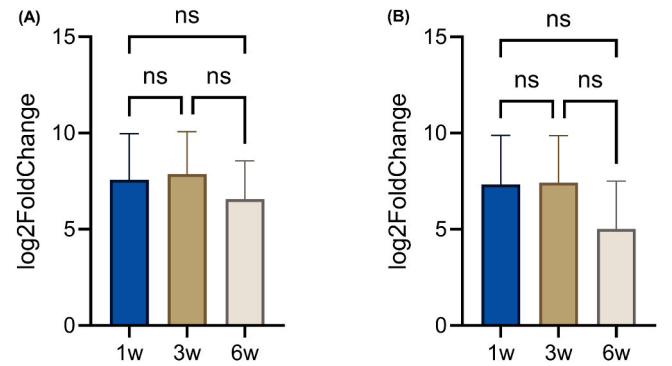


Fig. 5. Log2FC of T cell-related genes for (A) exhaustion and (B) co-stimulatory markers. ns, not significant; w, weeks. Genes presented in these figures are reflected in Table 3.

Simultaneously, the reduced expression of pattern recognition receptor (PRR) genes, such as nucleotide-binding oligomerization domain 1 (*NOD1*) and nucleotide-binding oligomerization domain 2 (*NOD2*), suggests diminished innate immune sensing and inflammatory priming [50]. However, these reductions were not significant at six weeks when

Table 3
Log2FC of T-cell exhaustion (and co-stimulation) markers.

Gene	1 week	3 weeks	6 weeks
<i>Exhaustion Markers</i>			
<i>PD-1</i>	9.277	10.89	10.99
<i>CTLA-4</i>	11.032	10.743	7.949
<i>LAG3</i>	6.358	6.103	4.229
<i>TIGIT</i>	-1.116	0	0
<i>BTLA</i>	12.314	11.637	9.657
<i>Co-stimulatory markers</i>			
<i>CD28</i>	5.482	5.738	3.168
<i>ICOS</i>	12.371	12.255	9.948
<i>CD27</i>	4.153	4.265	1.944

compared with earlier stages (one and three weeks).

Elevated expression of transcriptomic signatures associated with DCs such as *CD83*, *CD86*, *CD40*, *CCR2*, and *CCL19* at one week indicates a mature, pro-inflammatory DC phenotype actively engaged in antigen presentation and migration. High levels of *CCR5* and *CXCR4* at one week further supported active DC trafficking toward lymphoid tissues and tumor sites, while increased *IL-10* may reflect early regulatory feedback within an immunostimulatory context (Table S9). These transcriptional patterns suggest DC-mediated antigen presentation to T cells at 1 week. However, as the tumor progresses from three to six weeks, the potential DC-T cell interaction axis deteriorates sharply. For example, by six weeks, critical mediators such as *CCL19*, *CCL5*, and *CD40L* were lost, although groupwise comparisons involving DC-related genes between one and six weeks did not reach statistical significance ($p = 0.6$) (Fig. 6B, Table S9). However, the loss of these mediators between three and six weeks suggests a reduction in DC-mediated immune activation in late-stage tumors. This temporal remodeling likely reflects a tumor-driven shift toward immune evasion, which is increasingly recognized as a facilitator of metastatic progression [51–53]. This progressive decline of genes related to DC function is likely to affect the broader adaptive immune network, particularly B cells and macrophages, which depend on effective antigen presentation and cytokine signals from DCs and T helper cells. Mature DCs play a pivotal role in shaping B cell responses by presenting unprocessed antigens in lymphoid follicles, producing supportive cytokines (e.g., IL-6, IL-12), and licensing Tfh cell differentiation via co-stimulatory signals, such as CD40-CD40L [54, 55]. Gene expression associated with B-cell function remained relatively stable, with no significant differences observed between one, three, and six weeks. However, an overall reduction or loss of many transcriptomic signatures associated with B-cell function highlights a diminished B-cell-mediated immunity as the TME transitions to a more immunosuppressive state, consistent with tumor-driven immune escape mechanisms (Fig. 6C, Table S10).

In the immunologically dynamic context of TNBC, macrophages (also referred to as APCs) represent a central hub of immune regulation, toggling between pro-inflammatory (M1-like; anti-tumoral) and

immunosuppressive (M2-like; pro-tumoral) states [56, 57]. Transcriptomic signatures associated with classical M1-associated genes such as *IL-1 β* (7.79), *CCL5* (2.57), and *CXCL9* (10.22) were strongly expressed at one week and remained elevated or slightly increased at three and six weeks (Fig. 7A). Likewise, transcriptomic signatures related to chemokine receptors essential for monocyte and T cell recruitment, such as *CCR2* (15.16), *CCR5* (13.85), and *CCR7* (7.35), were also upregulated early, indicating an active immune infiltration landscape. Transcriptional regulator genes such as *STAT1* (2.16–2.81) and toll-like receptor 1 (*TLR1*) remained consistently expressed, pointing to stable innate immune activation (Fig. 7A).

In contrast, transcriptomic signatures associated with M2 polarization followed a distinct temporal trajectory. Colony-stimulating factor 1 (*CSF1*), a key M2-polarizing cytokine [58], remained significantly downregulated across all time points ($p < 0.0001$), while its receptor, CSF1 receptor (*CSF1R*), remained highly expressed ($p < 0.0001$), suggesting an environment that favors macrophage persistence but limits complete M2 differentiation. The M2-related cytokine *IL-13* increased at the three-week time point compared to the one-week time point (9.77–10.69), suggestive of transient Th2-mediated skewing [59] ($p < 0.01$). M2-related co-receptors, such as *IL-4R α* (3.47–3.11) and chemokine-like receptor 1 (*CMKLR1*; 8.60–8.69), remained stably expressed ($p < 0.05$), indicating persistent, but not escalating M2 signaling potential. Notably, the *ARG1* gene, a hallmark of metabolic immunosuppression, remained highly expressed across the time course ($p < 0.0001$), implying continuous arginine depletion and T cell suppression [60, 61]. Additional M2-related transcriptomic signatures, including *CD163*, *C3AR1*, and *ROR α* , also exhibited sustained expression ($p < 0.001$), reinforcing the presence of tolerogenic macrophages [62, 63]. Interestingly, *IL-10* expression declined from 6.24 at earlier stages to 3.53 at late stages ($p < 0.0001$), suggesting reduced immunosuppressive cytokine signaling in advanced tumors (Fig. 7B).

Cell-type deconvolution revealed dynamic changes in immune infiltrates, including APCs and T cells, over time (Fig. 8). For instance, at one week, tumors are enriched with macrophages, granulocyte-monocyte progenitors, and B cells, consistent with early innate responses to emerging tumor antigens. By three weeks, there is a marked increase in adaptive immune infiltration, with elevated scores for CD8⁺ and $\gamma\delta$ T cells, indicating a developing adaptive immune response. However, as the tumors progress to late stages (i.e., at six weeks), the TME shows a shift toward naïve B cells and immature T cells, and an enrichment of stem-like cells, which may reflect immune evasion and tumor progression (Table S3). These temporal trends from early innate immunity through adaptive activation to late dominance of immature B/T cells are consistent with established concepts of tumor-immune dynamics and the evolving cellular heterogeneity within tumors during progression and immune editing.

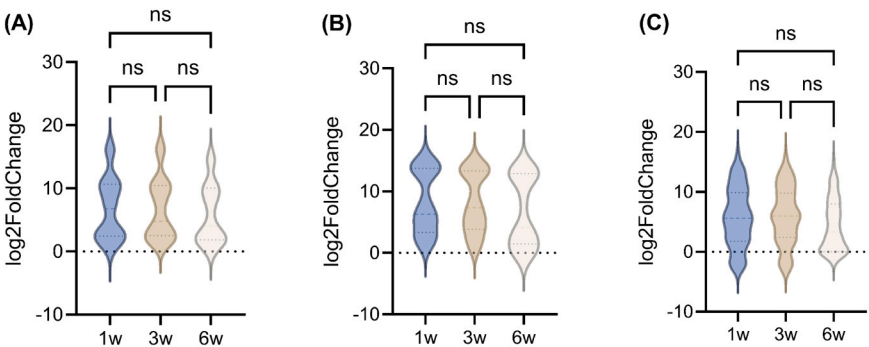


Fig. 6. Log2FC of (A) antigen processing/presentation cell (APC) genes, (B) dendritic cell genes, and (C) B cell genes. ns, not significant; w, weeks. Genes presented in these figures are reflected in Tables S8–S10.

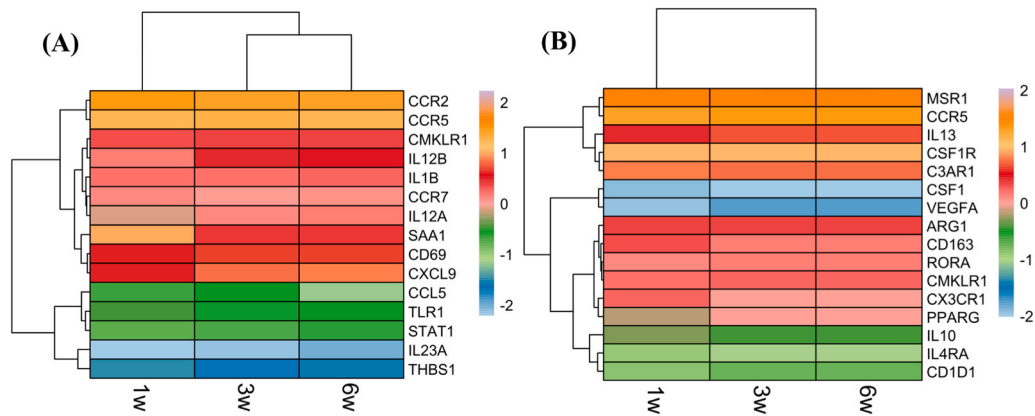


Fig. 7. Heatmap of macrophage-associated genes during 4T1 tumor growth. (A) M1-associated genes, and (B) M2-associated genes, where orange to red shades indicate upregulation and green to blue shades indicate downregulation. Gene expression levels are shown at the indicated time points of tumor progression, with the scale bar representing normalized values from -2 (lowest expression) to +2 (highest expression).

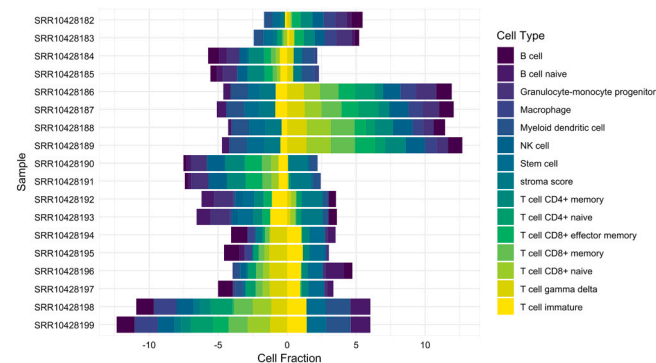


Fig. 8. Temporal changes in tumor immune cell composition at 1-, 3- and 6-Weeks.

4. Discussion

Our study demonstrates that T cells within the 4T1 TNBC TME undergo profound temporal remodeling during tumor progression, with implications for both tumor adaptation and therapeutic response. These results align with recent work showing that the TME of breast tumors undergoes continuous remodeling over four weeks, including increased infiltration of DCs that remains stable during the first two weeks; the absence of T helper cells and Tregs in week 2, followed by a peak in week 3 and a reduction in week 4; and a progressive increase in cytotoxic CD8⁺ T cells [64]. Consistent with the early monocyte/DC colonization phase, we observed robust upregulation of genes associated with T cell activation in early-stage tumors, characterized by elevated expression of transcriptomic signatures related to cytotoxic effector genes (*PRF1*, *GZMB*, and *IFN-γ*) and dynamic polarization, supported by expression of *IL-12/STAT4* and *IFN-γ*. As tumors progressed, however, this activation signature was gradually reduced, which is indicative of T cell dysfunction and exhaustion across T cell subsets.

While transcriptomic signatures related to exhaustion marker genes, such as *PD-1* and *CTLA-4*, remained stably expressed across tumor progression, co-stimulatory receptor genes, including *CD28* and *ICOS*, were significantly downregulated by late tumor stages. This combination of persistent expression of transcripts related to inhibitory signaling and diminished expression of transcripts associated with co-stimulation defines a state of terminal exhaustion, in which T cells retain inhibitory checkpoints but lose their capacity for effective activation and proliferation [47, 65, 66]. These findings suggest that the shift toward exhaustion is not merely a static phenotype but a temporally regulated process driven by sustained antigen exposure and cumulative

immunosuppressive signaling within the tumor milieu. Concurrently, analysis of cytokine and chemokine expression genes revealed transient peaks in Th1/Tc1 and Tfc-like activity at early and intermediate stages, followed by suppression of cytokine activity genes (*IL-5*, *IL-13*, and *IL-21*) at later stages of tumor growth, alongside sustained expression of immunosuppressive *ARG1* and *TGF-β* genes. These findings illustrate the temporal remodeling of T cell subsets and the reprogramming of immune cytokine networks that underlie the progression from immune activation to immune suppression.

APC dysfunction emerged as a parallel hallmark of immune decline. At one week, high expression of *CCR7*, *H2-Aa*, *H2-Ab1*, *CD83*, *CD86*, and *CD40* genes supports productive T cell priming. By six weeks, however, classical antigen presentation genes, chemokine genes (e.g., *CCL19* and *CCL5*), and co-stimulatory mediator genes, including *CD40L*, were markedly reduced, indicating a functional and progressive decline of DC-T cell interactions, accompanied by a decrease in the expression of transcripts associated with immunoregulatory molecules (*IDO1*, *IL-10*). These changes are consistent with tolerogenic reprogramming [67], limiting effective T cell priming and perpetuating exhaustion. Similarly, B cell pathway-related transcriptomic signatures associated with antigen processing and B cell activation declined over time, suggesting impaired humoral support for antitumor immunity. Together, these alterations indicate a systematic failure of adaptive immune coordination as TNBC progresses.

Macrophage transcriptional dynamics further illustrated the adaptive complexity of the tumor immune landscape. For example, early enrichment of proinflammatory M1-like genes (e.g., *IL-1β*) gave way to immunosuppressive states marked by increased *ARG1* and *TGF-β* genes. *ARG1*, in particular, acts as a central metabolic suppressor of T cell activity by depleting L-arginine, an amino acid essential for T cell proliferation and cytokine production [68]. Overall, macrophages maintained a mixed phenotype, e.g., M1-like cytokine gene (e.g., *CCL5*) initially upregulated and then completely disappeared at the late stage, while M2-associated marker genes (e.g., *ARG1*, *CD163*) were sustained across all three time points. These findings highlight the fact that immunosuppressive macrophages take over immunostimulatory macrophages over time during tumor progression, likely contributing to immune escape and tumor progression.

This study has limitations, as transcriptomic profiling alone cannot fully elucidate the epigenetic and metabolic mechanisms underlying dynamic reprogramming of T cells within the TME, and bulk RNA-seq cannot distinguish compositional shifts from true transcriptional plasticity. Another key limitation of this study is that, because bulk RNA-seq was used, the findings represent population-level transcriptional correlations rather than direct evidence of functional interactions or cell-state transitions at single-cell resolution. Thus, whole-genome sequencing,

single-cell transcriptomics, proteomics, integrative multi-omics approaches, and *in vitro/ex vivo* co-culture studies will be crucial for further elucidating how T cells and APCs are reprogrammed within the TME. Further orthogonal validations of our findings may include: (i) flow cytometry to quantify T cell subsets or exhaustion markers; (ii) immunohistochemistry to assess infiltration dynamics; (iii) TCR extraction from bulk RNA-seq for verification of repertoire changes; and (iv) gene set or pathway enrichment analysis. However, these data, taken together, delineate a temporal trajectory from early immune activation to late-stage exhaustion and immune collapse in TNBC. Furthermore, these observations emphasize that temporal remodeling of T cells within the TNBC TME is dynamic, shifting from initial activation toward progressive exhaustion, impaired antigen presentation, and immune suppression. This dynamic immune adaptation provides tumors with a dual advantage: evading early cytotoxic responses while creating conditions favorable for metastatic progression.

5. Conclusions

In summary, this study provides us with the following possible major interpretations. First, temporal mapping of transcriptomic signatures appears to be associated with T cells in TNBC *in vivo*. Unlike previous studies focused on tumor cell or macrophage plasticity, this study provides longitudinal, transcriptome-wide profiling of intratumoral T cells across early, intermediate, and late stages of TNBC (one-, three-, and six-weeks post-implantation). It reveals dynamic, stage-specific transcriptional shifts that redefine T cell identity and function over tumor progression. Second, discovery of progressive narrowing of TCR diversity and loss of unconventional T-cell subsets suggests a decline in T-cell-associated genes and a significant loss of TCR diversity over time. For example, reductions in NKT and $\gamma\delta$ T cell-related transcripts suggest selective immune exhaustion and loss of unconventional T-cell support in late TNBC. Third, the study identified temporally distinct T-cell polarization states. In this respect, we uncovered a sequential functional reprogramming of intratumoral T cells. For instance, Th1/Tc1 polarization via IL-12/STAT4 signaling (pro-inflammatory, cytotoxic phase) at one week, the emergence of cytotoxic follicular-like T cells (Tfc) linked to *IL-21* and *BCL6* expression at three weeks, and transition toward Tc17/Tc22-like states driven by *Ahr* and *IL-1 β* expression at six weeks is consistent with immune dysfunction. Fourth, the data offer new insight into likely APC-T cell interaction breakdown over time. For example, the study demonstrates a progressive decline of APC co-stimulatory capacity and reduced APC gene expression at late stages, linking T-cell dysfunction directly to a failure of T-cell/APC crosstalk. Finally, a persistent tolerogenic macrophage niche suggests that TNBC maintains a stable, immunosuppressive macrophage environment throughout tumor progression. This conclusion is supported by the persistent *ARG1* and M2-like gene expression, despite early macrophage signaling being M1-like.

Furthermore, by defining the transcriptional programs underlying T cell dysfunction and APC decline, this study highlights key vulnerabilities that could be exploited for therapeutic reprogramming. In fact, our findings underscore the importance of timing in therapeutic interventions. Optimizing the timing and targets of immunotherapy to counteract maladaptive T-cell temporal remodeling may be crucial for enhancing immunotherapy outcomes in TNBC. For example, our temporal analysis of the 4T1 model reveals a therapeutic window during the early-to-intermediate stages of tumor progression (weeks 1–3), based on several possible interpretations derived from the following transcriptomic signatures related to: (i) when T cells retain IL-12/STAT4-driven cytotoxic potential, (ii) Tfc-like IL-21/BCL6 programs begin to emerge, (iii) APC co-stimulatory capacity remains partially preserved, and (iv) the immunosuppressive macrophage niche has not yet fully stabilized. Within this window, strategies that enforce early co-stimulatory signaling (e.g., agonists for CD28 or ICOS), bolster IL-12/IL-21-responsive cytotoxic states, maintain TCR diversity, or prevent

the subsequent collapse of APC function are most likely to translate effectively to clinical settings. Conversely, by week six, the transcriptomic signatures associated with the dominance of ARG1-mediated metabolic suppression, loss of TCR diversity, and terminal T cell exhaustion limit the efficacy of immunotherapeutic intervention, underscoring the importance of early targeting. Future studies employing single-cell and spatial multi-omics analyses will be crucial for defining the mechanistic underpinnings of this immune plasticity and identifying actionable checkpoints for reprogramming the tumor-immune interface.

Author statement

We confirm that this manuscript is original and that this manuscript has not been previously published (besides pre-print) and has not been submitted for publication elsewhere while under consideration, and all authors have approved the content and submitted version of this manuscript. All authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest, and any potential conflict of interest has been disclosed.

Author contributions

M.I.: Performed data analysis, prepared figures and tables, and wrote the manuscript; R.H.N.: Molecular biology and cell signaling; S.H.H. and R.B.J.: Bioinformatics and data modeling; P.M.M.: Breast cancer genomics; B.L.H.: Preclinical cancer immunology; M.T.H.: Overall guidance to the first author; C.J.R.: 4T1 model system; M.D.T.: Evolutionary genomics; J.L.G.: Evolutionary biology and genomics; H.L.K.: Clinical cancer immunology; D.S.: Conceptualization, study outline, supervision, and data analysis. All authors edited the manuscript and approved the submission.

CRediT authorship contribution statement

Bryan L. Holloman: Writing – review & editing. **Perpetua M. Muganda:** Writing – review & editing. **Roshonda B. Jones:** Writing – review & editing. **Scott H. Harrison:** Writing – review & editing. **Robert H. Newman:** Writing – review & editing. **Dipongkor Saha:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Md. Iftehimul:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation. **Howard L. Kaufman:** Writing – review & editing. **Graves Jr. Joseph L.:** Writing – review & editing. **Misty D. Thomas:** Writing – review & editing. **Checo J. Rorie:** Writing – review & editing. **Muhammad T. Hossain:** Writing – review & editing.

Declaration of Competing Interest

All authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. H.L.K. is an employee of Ankyra Therapeutics and serves on the boards of Crighton Biosciences and Marengo Therapeutics. He is also an advisor to ImmVira, Tatum Biosciences, and Virogin.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.csbj.2025.12.022](https://doi.org/10.1016/j.csbj.2025.12.022).

Data availability

Data comprising sequence read counts and differentially expressed genes (DEGs) for the 1-, 3-, and 6-week periods are deposited under DOI: 10.5281/zenodo.16886936.

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